

Weekly Diary – Week 7

Internship Organization: Vikaspedia

Duration: 01/03/2026 – 07/03/2026

Objective of the Week

The primary aim for this week was to gain a clear understanding of the requirements involved in building a real-time collaborative editing system. The focus was on analyzing how such systems operate in practice and designing a foundational architecture that integrates both frontend and backend components. Special attention was given to planning the editor interface and establishing a strategy for real-time communication using WebSockets to ensure smooth and synchronized data exchange between users.

Date – 1 March

On the first day, the major focus was on studying and interpreting the project requirements related to implementing real-time collaborative editing functionality. This involved carefully examining how multiple users can simultaneously edit a shared document without conflicts. Systems such as Google Docs were analyzed to understand their working principles. Key concepts like real-time synchronization, operational efficiency, and maintaining low latency communication were explored in detail. This helped in building a conceptual foundation for the project.

Date – 2 March

The second day was dedicated to designing the overall system architecture. A structured plan was created that defined how the frontend, backend, and database would interact with each other. The frontend was planned using ReactJS due to its component-based structure and efficiency in building dynamic user interfaces. For real-time communication, WebSockets were selected as the primary technology to enable instant updates between users. The architecture design ensured scalability and smooth data flow across all components.

Date – 3 March

On this day, the initial setup of the frontend environment was completed. A new React project was created and configured properly. A basic user interface for the text editor was developed, including a simple layout with an editable text area. This served as the foundation for further enhancements and allowed testing of user input functionality. The goal was to ensure that users could interact with the editor in a smooth and intuitive way.

Date – 4 March

The work on the fourth day focused on enhancing the editor by implementing a rich text editing feature. Using available libraries, a more advanced editor was integrated that allowed users to not only type text but also apply formatting such as bold, italics, and other styles. This step significantly improved the usability of the application and made it closer to real-world collaborative editing tools. Proper testing was done to ensure that text input and formatting worked without errors.

Date – 5 March

On the fifth day, backend development was initiated using Spring Boot. The initial project structure was set up, focusing on creating a scalable and maintainable backend environment. Configuration for handling WebSocket connections was started, which is essential for enabling real-time communication between the client and server. This marked the beginning of integrating backend logic with the frontend interface.

Date – 6 March

The sixth day was mainly focused on understanding the working of the WebSocket protocol in depth. Concepts such as persistent connections, bidirectional communication, and real-time data transfer were studied. A basic WebSocket connection was successfully implemented between the client and the server. Testing was conducted to ensure that the connection was stable and messages could be transmitted effectively, confirming that the setup was working as expected.

Date – 7 March

On the final day of the week, basic integration testing between the frontend and backend was performed. The React-based editor interface was connected with the backend server to verify overall functionality. It was confirmed that the editor loaded correctly and the backend server operated without errors. Initial testing ensured that both components were functioning independently as well as together at a fundamental level.